Waupaca County

UW Extension

FALL/WINTER 2017-18

AGRICULTURE RESOURCE NEWSLETTER

Greg Blonde, UW-Extension Agriculture Agent...715-258-6230 (ext 2) or greg.blonde@ces.uwex.edu

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"I like the dreams of the future better than the history of the past"

- Thomas Jefferson -



Harvesting Late-Season Corn

According to Joe Lauer, Extension Corn Agronomist at UW-Madison, by late October to early November, field dry down rates usually drop to 1/4-1/2% per day, and by mid November, less than a 1/4% per day. By late November, drying rates will be negligible, if any drying occurs at all. So what are the options for dealing with wet corn? It's generally recommend that harvesting corn for dry grain storage should begin at about 24-25% kernel moisture. Allowing corn to field dry below 20% brings greater losses from stalk lodging, ear rots and wildlife damage. Be prepared for localized root lodging and stalk lodging that may slow harvest and contribute to yield losses. Here are recommended harvest options at different moisture levels:

- 33-35% (plant) = Silage harvest
- 29-32% (kernel) = High moisture morn ensiled
- 25-26% (kernel) = Ideal for combining
- 20-23% (kernel) = Ideal for picking
- < 20% (kernel) = Higher field losses, lower drying cost

To assess lodging potential use either the pinch test or the push test to check for stalk integrity. Conduct the pinch test by squeezing the second or third internode above the ground. If it collapses, stalk quality is compromised. The push test is performed by pushing a corn stalk to approximately 45 degree angle. If it breaks, stalk quality has been reduced. If 10 percent of the stalks tested show poor stalk quality or lodge at the root, then these fields should be harvested earlier.

To determine pre– and post-harvest ear loss, keep in mind one 3/4 pound ear in each 1/100 of an acre equals one bushel per acre loss (1/2 pound ear equals 2/3 bushel yield loss). To determine 1/100 of an acre, take the normal 1/1,000 acre distance and multiple by ten (x10). In 30" rows, 1/1000 of an acre is 17' 5", so 1/100 acre equals that distance across ten rows. For each ear in that area, there is one bushel per acre loss. Behind the combine, every two kernels per square foot on the ground equals one bushel per acre of machine loss. Typical harvest loss will range from 0.5 to 2.5 bushels per acre.

A two year study of field losses of corn left standing after October reported 3-5% loss in November, 22% in December and if the corn is allowed to stand in the field all winter, a loss of up to 40%. Balancing drying cost with time to finish harvesting, crop condition (lodging) and increased combine losses and lower corn prices makes the decision to delay corn harvest this fall especially tough.

Keep in mind it takes about 0.02 gallons of propane to remove one point of moisture from a bushel of corn, so to estimate drying costs for a high temperature dryer, multiply propane price by 0.02 and then multiply by the number of percentage points of moisture to be removed (ie. $1.25/gal propane with corn to be dried from 28 to 15% (13 points = $1.25 x 0.02 x 13 = $0.33 drying cost per bushel).

And don't forget, safety is no accident, especially during a late season harvest.

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Pricing Standing Corn

To evaluate options for standing corn, check out this smart phone app released in 2014. The app is free and available for Android smart phones and tablets on the Google Play store by searching for "HMSC" or go to:

https://play.google.com/store/apps/details?id=com.smartmapps.corncalculator&hl=en

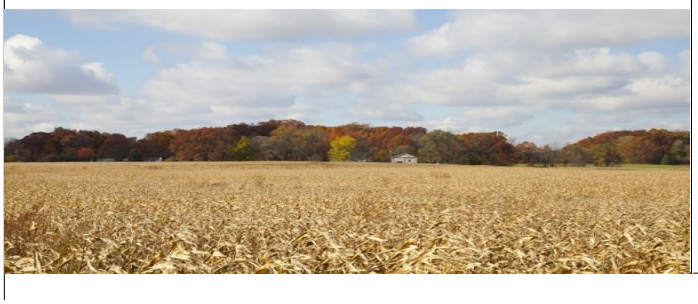
Farmers can use this app to help determine an equivalent value for wet shell corn when compared with a dry shell corn price (a link to current local elevator dry grain bid prices is also built into the app). The equivalent wet price is then calculated and displayed in both price per ton and price per bushel. Additional drying cost (propane or electric) can also be evaluated under the "expense" tab. Results can then be emailed from your phone to your computer, and/or emailed directly to others. Since 2014, more than 750 users have downloaded the app.

Managing Stress

It's important to recognize that it's impossible (in any job) to totally eliminate stress, but managing stress successfully is a realistic goal. Here's a few key points:

- Eat right
- Drink plenty of water
- Regular physical activity to elevate your heart rate
- Maintain regular sleep habits
- Make time to talk with family, friends & employees
- Keep your sense of humor
- Avoid unhealthy habits (tobacco, drugs & alcohol)

For more information on Farm Stress download "Farm Stress & Decision Making During Challenging Times" by John Shutske, Extension Specialist, UW Madison go to: https://fyi.uwex.edu/dairy/farm-stress-decision-making-during-challenging-times/



Upcoming UW-Extension Crop Management Programs

Soil, Water & Nutrient Mgmt Update Mtg

December 5th - Main Event, Cecil (8:30 - 11 AM)

- Cover Crops & Measuring Soil Health
- Soil Health and Crop Management Decisions
- Nitrogen sensors for Corn & Wheat
- Alfalfa S & K fertilization updates
- Precision Ag Technologies for Variable Rates
- WDATCP Erosion Vulnerability Assessment tool

Cost is $35/person . CEU credits (2.5) available. Register at least one week in advance by contacting the Shawano Co UWEX Office at 715-526-6136 or emailing Jamie Patton (jamie.patton@ces.uwex.edu).

Annual UW Agronomy Update Mtg

January 4th - 7:30 AM Liberty Hall, Kimberly

- Meet new UWEX Weed Specialist - Dr. Werle
- New Corn & Soybean Herbicides for 2018
- UW Forage / Grain Variety Trials
- Managing Corn Hybrid Selection
- High Input Corn vs. Economic Optimum Yield
- Revisiting Soybean Development & Management
- Cover Crops to Manage Weeds? Lessons Learned

Cost is $45/person (includes breakfast). CEU credits (3.0) available. Register at least one week in advance to avoid late fees by contacting Outagamie County UW-Extension (920-832-5119 or email Kevin Jarek (kevin.jarek@ces.uwex.edu).

Private Pesticide Applciator Training

Feb 27 (Tues) 9 AM Courthouse LL42
March 1 (Thur) 9:00 AM Manawa City Hall

Cost $30 per person. Those who need to re-certify or would like to become certified private pesticide applicators for field or vegetable crops can register by visiting the Waupaca County UWEX office at the courthouse in Waupaca. For an additional $5 ($35 total), the training manual can be mailed. Call (715-258-6231) or email Greg Blonde (greg.blonde@ces.uwex.edu) for more information.

Equivalent Price / Ton of High Moisture Shelled Corn*

Moisture % Wet	Dry Bushels	Price per Dry Bushel $2.90	$3.00	$3.10	$3.20	$3.30	$3.40	$3.50	$3.60	$3.70	$3.80
15.0	35.71	$103.57	$107.14	$110.71	$114.29	$117.86	$121.43	$125.00	$128.57	$132.14	$135.71
16.0	35.29	$101.51	$105.04	$108.56	$112.09	$115.62	$119.15	$122.68	$126.21	$129.74	$133.27
17.0	34.87	$99.46	$102.95	$106.44	$109.92	$113.41	$116.90	$120.38	$123.87	$127.36	$130.85
18.0	34.45	$97.44	$100.88	$104.33	$107.77	$111.22	$114.66	$118.11	$121.55	$125.00	$128.44
19.0	34.03	$95.43	$98.83	$102.24	$105.64	$109.04	$112.45	$115.85	$119.25	$122.66	$126.06
20.0	33.61	$93.45	$96.81	$100.17	$103.53	$106.89	$110.25	$113.61	$116.97	$120.34	$123.70
21.0	33.19	$91.48	$94.80	$98.12	$101.44	$104.76	$108.08	$111.40	$114.72	$118.04	$121.35
22.0	32.77	$89.54	$92.81	$96.09	$99.37	$102.65	$105.92	$109.20	$112.48	$115.75	$119.03
23.0	32.35	$87.61	$90.85	$94.08	$97.32	$100.55	$103.79	$107.02	$110.26	$113.49	$116.73
24.0	31.93	$85.71	$88.90	$92.09	$95.29	$98.48	$101.67	$104.87	$108.06	$111.25	$114.45
25.0	31.51	$83.82	$86.97	$90.13	$93.28	$96.43	$99.58	$102.73	$105.88	$109.03	$112.18
26.0	31.09	$81.96	$85.07	$88.18	$91.29	$94.40	$97.51	$100.62	$103.72	$106.83	$109.94
27.0	30.67	$80.12	$83.18	$86.25	$89.32	$92.38	$95.45	$98.52	$101.59	$104.65	$107.72
28.0	30.25	$78.29	$81.32	$84.34	$87.37	$90.39	$93.42	$96.44	$99.47	$102.49	$105.52
29.0	29.83	$76.49	$79.47	$82.46	$85.44	$88.42	$91.41	$94.39	$97.37	$100.35	$103.34
30.0	29.41	$74.71	$77.65	$80.59	$83.53	$86.47	$89.41	$92.35	$95.29	$98.24	$101.18
31.0	28.99	$72.94	$75.84	$78.74	$81.64	$84.54	$87.44	$90.34	$93.24	$96.14	$99.04
32.0	28.57	$71.20	$74.06	$76.91	$79.77	$82.63	$85.49	$88.34	$91.20	$94.06	$96.91
33.0	28.15	$69.48	$72.29	$75.11	$77.92	$80.74	$83.55	$86.37	$89.18	$92.00	$94.81
34.0	27.73	$67.77	$70.55	$73.32	$76.09	$78.87	$81.64	$84.41	$87.19	$89.96	$92.73
35.0	27.31	$66.09	$68.82	$71.55	$74.29	$77.02	$79.75	$82.48	$85.21	$87.94	$90.67
36.0	26.89	$64.43	$67.12	$69.81	$72.50	$75.19	$77.88	$80.56	$83.25	$85.94	$88.63
37.0	26.47	$62.79	$65.44	$68.08	$70.73	$73.38	$76.02	$78.67	$81.32	$83.96	$86.61
38.0	26.05	$61.17	$63.77	$66.38	$68.98	$71.59	$74.19	$76.80	$79.40	$82.01	$84.61
39.0	25.63	$59.56	$62.13	$64.69	$67.25	$69.82	$72.38	$74.94	$77.51	$80.07	$82.63
40.0	25.21	$57.98	$60.50	$63.03	$65.55	$68.07	$70.59	$73.11	$75.63	$78.15	$80.67
41.0	24.79	$56.42	$58.90	$61.38	$63.86	$66.34	$68.82	$71.30	$73.77	$76.25	$78.73

* If you must harvest this corn, subtract the harvesting costs; and if you can only use dry corn, subtract drying costs. Of course the pric you pay will be determined by supply and demand conditions in your area and negotiations between you and the seller, but unless there are special circumstances it should not be above the prices shown here.

Greg Blonde, UW-Extension Agriculture Agent...715-258-6230 (ext 2) or greg.blonde@ces.uwex.edu



2018 Cow College

FVTC Regional Center
Hwy 22/45 Clintonville, Wisconsin

January 9 *(1-3 PM)*

Using Genomic Testing to Improve Your Herd

Dr. Kent Weigel, *Extension Dairy Genetics Specialist, UW-Madison*

Dr. Weigel will discuss what we've learned about genomic testing in just the last few years to help improve dairy herd genetics and management. Is it really worth the effort and the cost?

Premium Beef Programs: An Option for Your Dairy Farm?

Dr. Victor Cabrera, *Extension Dairy Management Specialist, UW-Madison*

Dr. Cabrera will share economic analysis of cross-breeding low potential heifers and low producing dairy cows to beef bulls with superior carcass traits as an option for adding value to your bottom line.

January 16 *(1-3 PM)*

Feeding 2017 Forage & Grain Crops to Dairy Cows

Dr. Randy Shaver, *Extension Dairy Nutrition Specialist, UW-Madison*

Dr. John Goeser, *Rock River Lab Director of Research & Innovation, Watertown WI & Dairy Science Department Adjunct Assistant Professor, UW-Madison*

Dr. Goeser will review forage and grain quality from lab results in 2017, including energy, protein and fiber levels, as well as mycotoxins. Dr. Shaver will then provide an update and outlook on dairy nutrition and feeding strategies for 2018.

January 23 Farm Tour – Shawano County

Tour Locations Forthcoming

Lunch Speaker - Matt Akins, Extension Dairy Specialist - Updated Cost of Raising Dairy Replacements

Register for the sessions you plan to attend: ____ **January 9** ____ **January 16** ____ **January 23 (Farm Tour)**

Name(s) _____ Email_____

Address _____City _____ Zip _____ Phone _____

Cost is $5.00/day or $10.00 total for all three = $ _____ (Call or send check to UW-Extension by Fri. Jan 5)

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Zen Miller
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Appleton WI 54914
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Dairy Market Situation & Outlook

By Dr. Bob Cropp, Professor Emeritus
UW-Madison Extension Dairy Marketing Specialist

Higher cheese prices more than offsetting lower dry whey prices will result in an October Class III price near $16.60, up about $0.30 from $16.36 in September. Unless cheese prices rally more like prices did last year the Class III price for November and December will stay in the $16's and average about $16.25 for the year compared to $14.87 last year. Looking into 2018 Class III prices could stay in the high $15's for the first half of the year and reaching the $16's during the second half.

Relatively high stocks of cheese has dampened the increase in prices this fall. Dairy exports have supported cheese and butter prices. Cheese exports for August were 35% higher than a year ago and up 24% year-to-date. August butterfat exports were 177% higher than a year ago and 12% higher year-to-date. But, August nonfat dry milk/skim milk powder exports were 9% lower than a year ago, the second straight month of decline. August Dry whey exports were also lower than a year with a decline of 10%. Both nonfat dry milk/skim milk powder and dry whey are experiencing strong market competition for exports from the EU. Competition for markets will remain strong in 2018 as milk production is expected to increase in the two largest exporters, EU and New Zealand with also some milk production recovery in Argentina and Australia. Any major changes in trade agreements, in particular NAFTA could also impact exports. But, world demand is expected to increase as China and other major importers expand their imports of dairy products. This will help to absorb some of the increase in world milk production.

Milk prices for the remainder of this year and into 2018 will depend upon the level of milk production. USDA's report for September milk production showed the increase in milk slowed to 1.1% compared to increases of 2% for the previous two months. Milk cow numbers decline 4,000 head from August but were still 69,000 head or 0.7% higher than a year ago. The slowdown in milk production was due to milk per cow which was up just 0.3% from a year ago. USDA is forecasting 2018 milk production to increase 1.9% from this year, the result of 0.5% more cows and 1.4% more milk per cow. This is a lot of milk that will put downward pressure on milk prices. It would take higher than expected domestic sales of milk and dairy products or higher dairy exports to push 2018 milk prices higher than what is now forecasted.



Month	Class III	Class IV	Mailbox
Aug 2017	16.57	16.61	17.80
Sep 2017	16.36	15.86	17.50
Oct 2017	16.71	14.89	18.52
Nov 2017	16.54	14.15	18.26
Dec 2017	16.00	14.11	17.79
Jan 2018	15.70	14.23	17.38
Feb 2018	15.63	14.38	17.34
Mar 2018	15.48	14.54	17.24
Apr 2018	15.39	14.69	16.65
May 2018	15.50	14.84	16.77
Jun 2018	15.73	15.01	16.99
Jul 2018	15.97	15.21	17.07
Aug 2018	16.11	15.40	17.22
Sep 2018	16.18	15.60	17.31

heifer BLUEPRINT

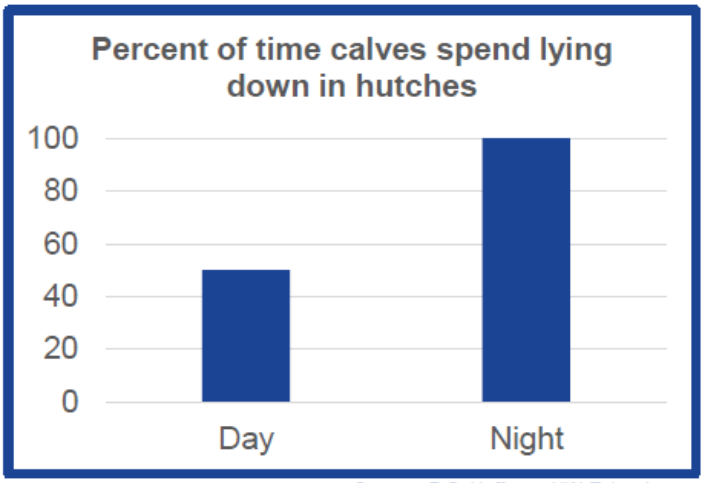
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Bedding Comfort



Providing an excellent environment is a key component in successfully raising calves. For calf housing, the environment includes the calf's resting space, feeding and walking surface.

Research shows young calves spend most of their time lying down, making the quality of the resting surface critical. A clean, dry resting surface ensures a hair coat that helps insulates the calf against the cold ground, low air temperatures, and sudden changes in temperature.



Source: P.C. Hoffman, UW-Extension

Knee Test for Comfort & Cleanliness

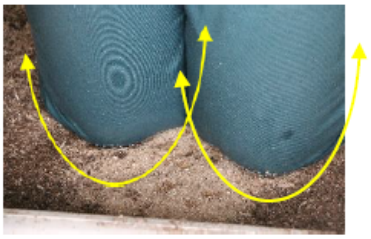


Photo courtesy of J. Patton

Test 1: Kneel on bedded surface, rocking back and forth to see if pressure is comfortable to knee. Does it feel comfortable?

- ✓ If yes, proceed to Test 2
- ✓ If no, add additional bedding. Repeat Test 1.

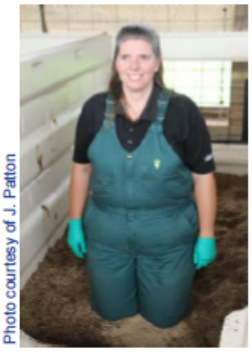


Photo courtesy of J. Patton

Test 2: From a standing position, bend your knees and drop quickly to the bedded surface. Does the impact feel comfortable?

- ✓ If yes, proceed to Test 3.
- ✓ If no, add additional bedding. Repeat Test 2.



Photo courtesy of J. Patton

Test 3: Check you knees. Are they dry or are they wet?

- ✓ If dry, the test is complete. Calf bedding acceptable.
- ✓ If wet, clean pen and/or provide additional dry bedding. Repeat Test 3.

Originally developed by UW-Extension Oconto County Agriculture Agent Sarah Mills-Lloyd & Fond du Lac County Dairy & Livestock Agent Tina Kohlman for the Nestlé Dairy Farming Institute Curriculum, and adapted for UW-Extension Dairy Team Heifer Blueprints. Photo credits: Shawano County Agriculture Agent Jamie Patton.
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heifer BLUEPRINT

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Bedding Comfort



"A calf can stand a good deal of cold weather if it is dry and protected from drafts."

-Agricultural Experiment Station Circular, July 1931



Photo courtesy of T. Kohlman

Cold Weather Comfort

- A dry place to lie
- Plenty of clean, dry bedding
- Plenty of fresh air
- No cold air drafts
- Minimum frost or condensation during very cold weather

Ideal Bedding



- Provide a good base to:
  - Soak up liquid
  - Provide insulation
  - Allow nesting
- 20-25 pounds of bedding per calf
- 2-3 pounds additional bedding per day

Nesting Scores for Cold Weather



Photo courtesy of S. Mills-Lloyd

Nesting Score 1
Legs are entirely visible when lying down.



Photo courtesy of S. Mills-Lloyd

Nesting Score 2
Legs are partially visible when lying down.



Photo courtesy of A. Bjurstrom

Nesting Score 3
Legs are not visible when laying down.

Originally developed by UW-Extension Oconto County Agriculture Agent Sarah Mills-Lloyd & Fond du Lac County Dairy & Livestock Agent Tina Kohlman for the Nestlé Dairy Farming Institute Curriculum, and adapted for UW-Extension Dairy Team Heifer Blueprints.
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Non-Profit Organization
U.S. Postal Paid
Waupaca, WI 54981
Permit No. 3

Upcoming Events:

December 5
Soil, Water, NM Update
Main Event, Cecil 8:30 –11 AM

December 12
UW Discovery Farm Conf
Glacier Canyon, WI Dells

January 4
UW Agronomy Update
Liberty Hall, Kimberly

January 9, 16, 23
Cow College, Clintonville

February 5
UWEX/DNR CAFO Update
Crystal Falls, New London

February 27
Pesticide Applicator Training
Courthouse, Waupaca

March 1
Pesticide Applicator Training
Manawa City Hall

Stay Connected with Local & Statewide Extension Resources



Stay current with the latest UW Extension dairy updates through the UW Extension Dairy Team website (http://fyi.uwex.edu/dairy/). Sign up for automatic email notices when new posting occur on topics related to:

- Ag Safety & Health
- Animal Health & Wellbeing
- Dairy Replacements
- Facilities & Modernization
- Financial Management
- Milk Quality & Milking Systems
- Feeding & Nutrition
- Human Resource Management
- Reproduction/Genetics
- Youth / Apps & More

Also, stay connected with timely updates from the Waupaca County UW-Extension Office through Facebook and Twitter postings at:

Facebook: facebook.com/WaupacaCountyUWEX

Twitter: @uwexwaupacaco

For more on these or other web based UW-Extension resources, including updating your contact information to continue receiving this newsletter, call Dana Nelson at 715-258-6231 or by email at dana.nelson@ces.uwex.edu.

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